Walkthrough for “Is_this_just_a.pdf” (Steganography)

1. Challengers will find the file “Is_this_just_a.pdf” in a folder named “guard_room_3a”.

```
┌──(root㉿kali)-[/home/kali/guard_room_3a]
└─# la
Is_this_just_a.pdf
```

2. Open with any PDF reader

Congratulation? The flag for guard_room_3a is: PrisonCTF{team2No1}

They will see a FAKE flag, this fake flag is actually the key used to extract the real flag.

3. The challenger needs to use tools such as binwalk to analyze the file structure

```
┌──(root㉿kali)-[/home/kali/guard_room_3a]
└─# binwalk Is_this_just_a.pdf

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
0             0x0             PDF document, version: "1.4"
194           0xC2            Zlib compressed data, default compression
1571          0x623           Zlib compressed data, default compression
18382         0x47CE          Zlib compressed data, default compression
19287         0x4B57          JPEG image data, JFIF standard 1.01
```

There is actually a JPG file embedded at the end of the PDF file.

4. Extract hidden files

```
┌──(root㉿kali)-[/home/kali/guard_room_3a]
└─# foremost Is_this_just_a.pdf
Processing: Is_this_just_a.pdf
|*|

┌──(root㉿kali)-[/home/kali/guard_room_3a]
└─# cd output

┌──(root㉿kali)-[/home/kali/guard_room_3a/output]
└─# ls
audit.txt  jpg  pdf

┌──(root㉿kali)-[/home/kali/guard_room_3a/output]
└─# cd jpg

┌──(root㉿kali)-[/home/kali/guard_room_3a/output/jpg]
└─# ls
00000037.jpg

┌──(root㉿kali)-[/home/kali/guard_room_3a/output/jpg]
└─# xdg-open 00000037.jpg

┌──(root㉿kali)-[/home/kali/guard_room_3a/output/jpg]
└─#
```

Note: There are many methods and tools available for extracting files; this walkthrough demonstrates how to use “foremost” for extraction. Since we determined the starting offset of the JPG file in the previous step, we can also use `dd` to manually extract the JPG.
Example: *dd if=Is_this_just_a.pdf of=extracted.jpg bs=1 skip=19287*

Eventually, the challenger will see the image below and realize that this is a steganography challenge.

5. Use steghide to extract the hidden content, the password is provided in the previous flag: team2No1, and obtain the actual flag file

```
┌──(root㉿kali)-[/home/kali/guard_room_3a/output/jpg]
└─# steghide extract -sf 00000037.jpg
Enter passphrase:
wrote extracted data to "flag.txt".

┌──(root㉿kali)-[/home/kali/guard_room_3a/output/jpg]
└─# cat flag.txt
PrisonCTF{feronia_and_milly_20010430}
```